THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A right circular cylinder has base radius 7 cm and height 5 cm. The curved surface area of the cylinder is: 1

(a) 528 cm² **(b)** 440 cm² **(c)** 770 cm² **(d)** 220 cm²

2. A right circular cylinder has base radius 7 cm and height 10 cm. The volume of the cylinder is: 1

(a) 3080 cm³ **(b)** 748 cm³ **(c)** 1540 cm³ **(d)** 440 cm³

3. A right circular cone has base radius 7 cm and height 24 cm. The slant height of the cone is: 1

(a) 25 cm **(b)** 15.5 cm **(c)** 23 cm **(d)** 31 cm

4. A right circular cone has base radius 7 cm and height 24 cm. The curved surface area of the cone is: 1

(a) 704 cm² **(b)** 528 cm² **(c)** 550 cm² **(d)** 154 cm²

5. A solid sphere has radius 7 cm. The surface area of the sphere is: 1

(a) 308 cm² **(b)** 462 cm² **(c)** 616 cm² **(d)** 154 cm²

6. A solid sphere has radius 21 cm. The volume of the sphere is: 1

(a) 29106 cm³ **(b)** 38808 cm³ **(c)** 19404 cm³ **(d)** 5544 cm³

7. A solid hemisphere has radius 7 cm. The total surface area of the hemisphere is: 1

(a) 462 cm² **(b)** 616 cm² **(c)** 308 cm² **(d)** 154 cm²

8. A right circular cylinder has base radius 14 cm and height 10 cm. The curved surface area of the cylinder is: 1

(a) 6160 cm² **(b)** 880 cm² **(c)** 1760 cm² **(d)** 2112 cm²

9. A right circular cylinder has base radius 14 cm and height 5 cm. The volume of the cylinder is: 1

(a) 1672 cm³ **(b)** 6160 cm³ **(c)** 440 cm³ **(d)** 3080 cm³

10. A right circular cone has base radius 14 cm and height 48 cm. The slant height of the cone is: 1

(a) 62 cm **(b)** 50 cm **(c)** 31 cm **(d)** 47 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A right circular cylinder has base radius 7 cm and height 10 cm. Find the total surface area of the cylinder. 2

12. A solid hemisphere has radius 10.5 cm. Find the volume of the hemisphere. 2

13. A cuboid measures 8 cm by 5 cm by 3 cm. Find the total surface area of the cuboid. 2

14. A right circular cylinder has base radius 14 cm and height 6 cm. Find the total surface area of the cylinder. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A right circular cone has base radius 7 cm and height 24 cm. Find the volume of the cone. 3

16. A right circular cone has base radius 7 cm and height 24 cm. Find the total surface area of the cone. 3

17. The total surface area of a cube is 54 cm^2. Find the volume of the cube. 3

OR

A right circular cone has base radius 14 cm and height 48 cm. Find the volume of the cone. 3

18. A right circular cone has base radius 14 cm and height 48 cm. Find the total surface area of the cone. 3

OR

The total surface area of a cube is 96 cm^2. Find the volume of the cube. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A solid metal cuboid 24 cm by 12 cm by 6 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 5

OR

A solid metallic sphere of radius 6 cm is melted and recast into small solid spheres of radius 2 cm. Find the number of small spheres obtained. 5

20. A solid metal cuboid 24 cm by 12 cm by 12 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. 5

OR

A solid metallic sphere of radius 9 cm is melted and recast into small solid spheres of radius 3 cm. Find the number of small spheres obtained. 5

END OF QUESTION PAPER Total: 40 Marks

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